An Introduction to the cov vs pp problem

Lecture 1: A Lore Dump

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Vocabulary

Ideal Language

Definition

Let J be an ideal over a set A.

- $J^+ = \{X \subseteq A : X \notin J\}$ is the collection of J-positive subsets of A.
- $J^* = \{X \subseteq A : A \setminus X \in J\}$ denotes the dual filter.
- $J \restriction B = \{X \subseteq A : X \cap B \in J\}$
- $J + B = J \restriction (A \setminus B)$

Comparisons modulo an ideal

Definition

For ordinal-valued functions with domain A

$$f <_J g \iff \{a \in A : g(a) \leq f(a)\} \in J,$$

$$f \leq_J g \iff \{a \in A : g(a) < f(a)\} \in J,$$

The definitions of $=_J$ and $\neq_J$ are similar.

Reduced Products

Our basic object of study will be reduced products $\prod A/J$ for A a set of regular cardinals and J an ideal on A.

We usually assume that A is *progressive*, which means $|A| < |\min(A)|$.

We may also consider indexed products $\prod_{i\in\Gamma}\lambda_i$ where Γ is some index set and λ_i is a regular cardinal. The reduced product makes sense if J is an ideal on Γ, and we say that this is a *small product* if $|\Gamma| < \mathbf{min}\{\lambda_i : i \in \Gamma\}$. These approaches are equivalent.

True cofinality

Let A be a progressive set of regular cardinals, and let J be a proper ideal on A. We say

$$\mathrm{tcf}(\prod A/J) = \kappa$$

if κ is regular and there is a sequence $\langle f_\alpha : \alpha < \kappa \rangle \subseteq \prod A$ such that

$$\alpha < \beta < \kappa \quad \Longrightarrow \quad f_\alpha <_J f_\beta,$$

and the sequence is cofinal:

$$(\forall g \in \prod A)(\exists \alpha < \kappa)\, g <_J f_\alpha.$$

$\mathrm{pcf}(A)$

If A is a progressive set of regular cardinals, then $\mathrm{pcf}(A)$ is the set of all cardinals $\mathrm{tcf}(\prod A/J)$ where J is an ideal on A and the true cofinality exists.

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Most important for us is that $\max \mathrm{pcf}(A)$ always exists, and it is equal to the cofinality of $(\prod A, <)$.

Big Picture

If λ is singular, there is a connection between

$$[\lambda]^{<\lambda} \qquad \text{and} \qquad \prod(\lambda \cap \mathrm{Reg}).$$

Questions about the former structure can be translated into questions about the latter, where pcf theory can be used to analyze the situation.

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How about an example?

Pseudopowers

Representability

Let λ be a singular cardinal. A cardinal κ is representable at λ if there are a set A and an ideal J such that

$$
\begin{aligned}
&A \text{ is cofinal in } \lambda \cap \mathrm{Reg},\\
&|A| = \mathsf{cf}(\lambda),\\
&J \text{ is an ideal on } A \text{ extending } J_A^{\mathrm{bd}},\\
&\kappa = \mathsf{tcf}\left(\prod A/J\right).
\end{aligned}
$$

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The Pseudopower Function

- $\mathrm{PP}(\lambda) = \{\kappa : \kappa \text{ is representable at } \lambda\}$
- $\mathrm{pp}(\lambda) = \sup \mathrm{PP}(\lambda)$
- Why is λ^+ representable at λ?
- If $A = \{\aleph_n : 1 \leq n < \omega\}$, show that $\mathrm{PP}(\aleph_\omega) = \mathrm{pcf}(A) \setminus A$ and $\mathrm{pp}(\aleph_\omega) = \max \mathrm{pcf}(A)$.

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There is some $\delta < \lambda$ such that $\mathsf{pp}(\lambda)$ is the least cardinal η with the property that, whenever $A \subseteq (\delta, \lambda) \cap \mathrm{Reg}$ and $|A| = \mathsf{cf}(\lambda)$, we have $\max \mathsf{pcf}(A) \leq \eta$.

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Can we prove it?

Note

Note that if λ is not a fixed point, then for all sufficiently large $\delta < \lambda$ the set $A = (\delta, \lambda) \cap \mathrm{Reg}$ is a progressive interval of regular cardinals, and we can apply pcf analysis in this setting. This is one reason we know so much more about cardinal arithmetic at non fixed points, and why getting information when $\lambda = \aleph_\lambda$ is more difficult.

Restricting the ideals

Assume $\aleph_0 \leq \sigma = \mathrm{cf}(\sigma) \leq \mathrm{cf}(\mu) < \theta < \mu$.

- $\Gamma(\theta, \sigma)$ is the class of proper σ-complete ideals on sets of cardinality $< \theta$.
- $\mathrm{PP}_{\Gamma(\theta,\sigma)}(\mu)$ consists of all regular cardinals κ such that

$$\kappa = \mathrm{tcf}\left(\prod A/J\right),$$

where

$$A \subseteq \mu \cap \mathrm{Reg}, \qquad \sup A = \mu, \qquad |A| < \theta,$$

and $J \in \Gamma(\theta, \sigma)$ is an ideal on A extending J_A^{bd}.

Covering Numbers

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The covering number $\mathsf{cov}(\lambda, \kappa, \theta, \sigma)$ is the least size of a family

$$\mathcal{X} \subseteq [\lambda]^{<\kappa}$$

such that every $a \in [\lambda]^{<\theta}$ is contained in the union of fewer than σ members of $\mathcal{X}$.

Easy Facts

For fixed values of the other parameters,

- $\mathsf{cov}(\lambda, \kappa, \theta, \sigma)$ is weakly decreasing as κ or σ increases,
- $\mathsf{cov}(\lambda, \kappa, \theta, \sigma)$ is weakly increasing as λ or θ increases.

Identities and a pcf Bound

Two easy observations:

$$\mathrm{cov}(\lambda, \kappa, \theta, \aleph_0) = \mathrm{cov}(\lambda, \kappa, \theta, 2),$$

$$\mathrm{cov}(\lambda, \kappa^+, \kappa^+, 2) = \mathrm{cof}\big([\lambda]^\kappa, \subseteq\big).$$

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One not as easy:

If $\aleph_0 \leq \sigma \leq \mathrm{cf}(\lambda) < \theta < \lambda$, then

$$\mathrm{pp}_{\Gamma(\theta,\sigma)}(\lambda) \leq \mathrm{cov}(\lambda, \lambda, \theta, \sigma).$$

The cov vs. pp Theorem

If $\aleph_0 < \sigma = \mathrm{cf}(\sigma) \leq \mathrm{cf}(\lambda) < \theta < \lambda$ then

$$\mathrm{pp}_{\Gamma(\theta,\sigma)}(\lambda) = \mathrm{cov}(\lambda, \lambda, \theta, \sigma)$$

.

The general cov vs. pp theorem

Suppose as usual $\lambda \geq \kappa \geq \theta > \sigma > \aleph_0$ where σ is regular, and put $\Gamma = \Gamma(\theta, \sigma)$.

Theorem (Shelah)

$$\begin{aligned} &\mathrm{cov}(\lambda, \kappa, \theta, \sigma) + \lambda \\ &\quad = \sup_{\substack{\kappa \leq \mu \leq \lambda \\ \sigma \leq \mathrm{cf}(\mu) < \theta}} \mathrm{pp}_\Gamma(\mu) + \lambda. \end{aligned}$$

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If $\kappa < \lambda$, then the $+\lambda$ on the left can be omitted.

Non-fixed points

If λ is singular and $\lambda < \aleph_\lambda$, then for every κ with $\mathsf{cf}(\lambda) \leq \kappa < \lambda$ we have

$$\mathsf{pp}_\kappa(\lambda) =^+ \mathsf{cov}(\lambda, \lambda, \kappa^+, 2).$$

In particular,

$$\lambda \text{ singular with } \lambda < \aleph_\lambda \Longrightarrow pp(\lambda) =^+ \mathsf{cov}(\lambda, \lambda, (\mathsf{cf}\ \lambda)^+, 2).$$

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Moti showed us that the answer to these questions is yes below 0-pistol.